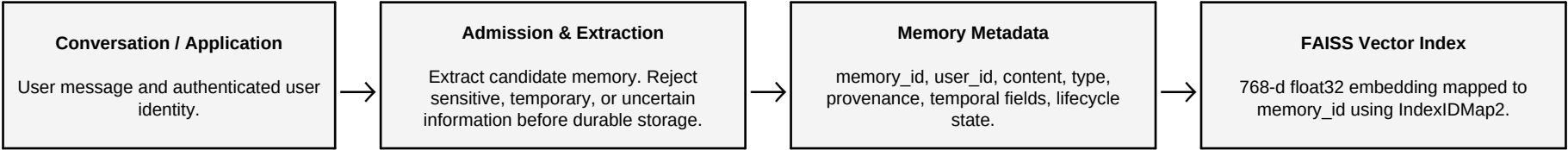


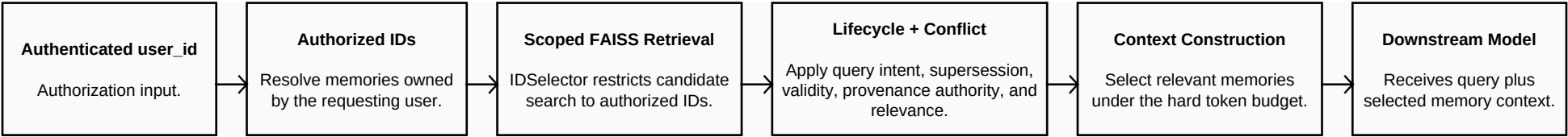
D4 - Conversational Memory System Architecture

Logical architecture consolidated from the D4 design decisions and experiments.

Memory Creation and Storage



Online Retrieval Path



Lifecycle and Privacy Controls

Supersession Newer explicit state can replace a specifically identified older memory; history is preserved.	Expiration Only explicit valid_until boundaries expire memory; no generic age-based decay.	Forgetting Logical exclusion is immediate. Physical FAISS deletion is attempted but is not the correctness boundary.	No-memory If no candidate reaches the relevance threshold, return no memory.
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D4 - Engineering Boundaries

Area	D4 position
Scale	Shared IndexFlatIP + IndexIDMap2 was successfully tested at 100K and 1M memories. The 5M index build failed with std::bad_alloc in the development environment.
Isolation	user_id is a hard authorization boundary. Authorized memory IDs are resolved before semantic candidate retrieval and enforced with FAISS IDSelector.
Current vs historical state	Superseded memories are excluded from current-state retrieval but may be recovered for explicit historical queries. Forgotten memories remain excluded.
Context budget	Memory context is selected under a hard token budget with response tokens reserved.
Recovery	Deletion failure behavior was tested. Complete persistent crash/restart recovery and source-of-truth design remain open.
API boundary	Logical store, retrieve, historical-retrieve, and forget operations are defined. Production transport, authentication, and deployment infrastructure are not implemented in D4.
Observability	D4 has experiment-level diagnostics and evaluation evidence, but not a unified production observability system.

Architecture boundary: This PDF describes the D4 logical architecture established by the project evidence. Production persistence, crash recovery, authentication infrastructure, deployment topology, and monitoring remain explicit engineering gaps rather than assumed capabilities.